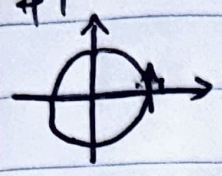
# HW#6

## #1



a) $a = \frac{v^2}{r}$

$= \frac{100^2}{50}$

$= 200000\ km/h^2 = 1.57g$

b) $a = \frac{v^2}{r}$

$= \frac{1500000}{5000}$

$= 300\ m/h^2 =$

c) $V = \frac{2\pi r}{T}$

$= \frac{2\pi \cdot 1.5m}{.5s}$

$= 18.849\ m/s$

$a = \frac{v^2}{r}$

$= \frac{18.849\ m/s^2}{1.5m}$

$= 236.871\ m/s$

d) $V = \frac{2\pi r}{T}$

$= \frac{2\pi \cdot 6in}{1/33.1\ min}$

$= 1247.841\ in/min$

$a = \frac{v^2}{r}$

$= \frac{1247.841\ in/min^2}{6in}$

$= 259{,}517\ in/min^2$

e) $V = \frac{2\pi r}{T}$

$= \frac{2\pi \cdot 15cm}{1/30{,}000\ min}$

$= 28{,}274\ m/min$

$a = \frac{v^2}{r}$

$= \frac{28{,}274\ m/min^2}{.15m}$

$= 5\,329\,586\,376.6\ m/min^2$

## #2

$g = a = \frac{v^2}{r}$

$V = t \cdot d$

$t = \frac{d}{V}$

$g = a = \frac{v^2}{r}$

$g = \frac{v^2}{r}$

$gr = v^2$

$V = \sqrt{gr}$

$V = \frac{d}{t}$

$Vt = d$

$t = \frac{d}{V}$

$t = \frac{d}{\sqrt{gr}}$

$t = \frac{d}{V}$

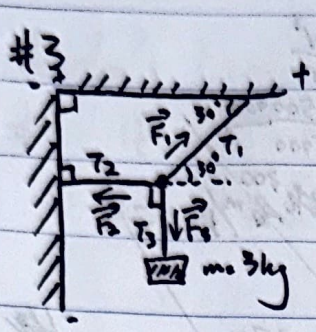
$= \frac{\text{Earth equator}}{\sqrt{gr}}$

$= \frac{40075\ km}{\sqrt{9.81\ m/s^2 \cdot 6378\ km}}$

$= \frac{40075\ km}{62.56818\ km^2/s}$

$= 640.501\ s$

#3




$\therefore \vec{F_1} + \vec{F_2} + \vec{F_3} = 0$

$\vec{F_3} = mg$

$= 3kg \cdot -9.81 m/s^2$

$= -29.43 \, kg \, m/s^2$

$\therefore \vec{F_1} + \vec{F_2} = 29.43 \, kg \, m/s^2$

$\because T_2$ is horizontal

$\therefore \vec{F_2} = (x\hat{i}, 0\hat{j})$

$\because$ the block is suspended

$\therefore T_2$ is perpendicular to $T_3$

$\therefore T_3$ is vertical

$\therefore \vec{F_3} = (0\hat{i}, -29.43 kg \, m/s^2)$

$\cancel{\vec{F_1}\hat{i} + \vec{F_2}\hat{i}}$

$F_1\hat{j} + F_2\hat{j} + F_3\hat{j} = 0$

$F_1\hat{j} + 0 + -29.43 \frac{kg \cdot m}{s^2} = 0$

$F_1\hat{j} = 29.43 \, kg \, m/s^2$

$F_1\hat{i} + F_2\hat{i} + F_3\hat{i} = 0$

$50.97 kg \, m/s^2 + F_2\hat{i} + 0 = 0$

$F_2\hat{i} = -50.97 kg \, m/s^2$

$F_1\hat{j} = F_1 \sin 30°$

$F_1 = \frac{F_1\hat{j}}{\sin 30°}$

$= \frac{29.43}{\sin 30°} kg \, m/s^2$

$= 58.86 \, kg \, m/s^2$

$\vec{F_1} = (50.97 kg \, m/s^2 \hat{i}, 29.43 kg \, m/s^2 \hat{j})$

$\vec{F_2} = (-50.97 kg \, m/s^2 \hat{i}, 0\hat{j})$

$F_1\hat{i} = F \cos 30°$

$= 58.86 \cdot \cos 30° \, kg \, m/s^2$

$= 50.97 \, kg \, m/s^2$

#4

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$\vec{F}_1 = (\overset{(12\cdot\cos 30^\circ\hat{i}}{\cancel{\cos 30^\circ}}, 12\sin 30^\circ\hat{j})$$

$$= (10.39N\hat{i}, 6N\hat{j})$$

$$\vec{F}_2 = (16\cdot\cos 145^\circ\hat{i}, 16\sin 145^\circ\hat{j})$$

$$= (-13.11N\hat{i}, 9.18N\hat{j})$$

$$\vec{F}_3 = (7\cos 245^\circ\hat{i}, 7\sin 245^\circ\hat{j})$$

$$= (-2.96N\hat{i}, -6.34N\hat{j})$$

$$\vec{F}_{net} = (F_1\hat{i} + F_2\hat{i} + F_3\hat{i}, F_1\hat{j} + F_2\hat{j} + F_3\hat{j})$$

$$= [(10.39N + -13.11N + -2.96N)\hat{i}, (6N + 9.18N + -6.34N)\hat{j}]$$

$$= (-5.68N\hat{i}, 8.84N\hat{j})$$

#5

a) $F = ma$

Given: $m = 1225\ kg$,
$V = 96.0\ km/h$, $V_0 = 0\ km/h$,
$t = 10\ s$

$V = V_0 + at$

$a = \frac{V - V_0}{t}$

$F = m\frac{V - V_0}{t}$

$= 1225\ kg\ \frac{96.0\ km/h - 0}{10\ s}$

$= 1225\ kg\ \frac{96.0\ km/h}{2.778 \times 10^{-3}\ h}$

$= 1225\ kg \cdot 34560\ km/h^2$

$= 4.2336 \times 10^7\ kg\ km/h^2$

b) $F = ma$

Given: $V = 0\ km/h$, $V_0 = 112\ km/h$,
$d = 64.0\ m$, $d_0 = 0\ m$,
$m = 1225\ kg$

$\cancel{a = \frac{V - V_0}{t}}$

$V^2 = V_0^2 + 2a(d - d_0)$

$a = \frac{V^2 - V_0^2}{2(d - d_0)}$

$F = m\frac{V^2 - V_0^2}{2(d - d_0)}$

$= 1225\ kg\ \frac{0^2 - 112\ km/h^2}{2(64.0\ m - 0)}$

$= 1225\ kg \cdot -\frac{12544\ km^2/h^2}{.128\ km}$

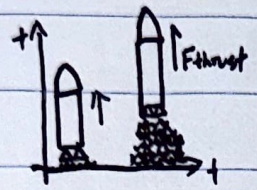
$= 1225\ kg \cdot -98000\ km/h^2$

$= -1.2005 \times 10^8\ km/h^2$

#6

a) 12.0 N  b) 24.0 N  c) 12.0 N

#7




Given: $\vec{F}_{thrust} = +2.00\times10^5\,N$, $m = 1.40\times10^4\,kg$, $t = 1\,min = 60\,s$, $\vec{F}_g = -m\,9.81\,m/s^2$

$$\vec{F}_{net} = \vec{F}_{thrust} + \vec{F}_g$$
$$= 2.00\times10^5\,N + -1.40\times10^4\,kg \cdot 9.81\,m/s^2$$
$$= 2.00\times10^5\,N - 1.37\times10^5\,N$$
$$= .63\times10^5\,N$$

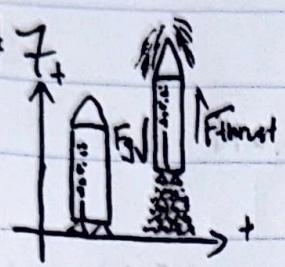
$$\vec{F}_{net} = m\vec{a}$$
$$\vec{a} = \frac{\vec{F}_{net}}{m} = \frac{\vec{F}_{thrust} + \vec{F}_g}{m}$$

$$V = V_0 + at$$
$$V = V_0 + \frac{\vec{F}_{thrust} + \vec{F}_g}{m}\,t$$

#4

$$V = V_0 + \frac{\vec{F}_{thrust} + \vec{F}_g}{m}\,t$$
$$= 0\,m/s + \frac{2.00\times10^5\,N + -1.40\times10^4\,kg \cdot 9.81\,m/s^2}{1.40\times10^4\,kg} \cdot 60\,s$$
$$= 4.5\times10^4\,m/s^2 \cdot 60\,s$$
$$= 2.7\times10^5\,m/s$$

#7




Given: $\vec{F}_{thrust} = 2.00 \times 10^5 N$, $\vec{F}_g = mg$

$m = 1.40 \times 10^4 kg$, $t = 1 min = 60 s$

$x_0 = 0 m$, $V_0 = 0 m/s$

$$\vec{F}_{net} = \vec{F}_{thrust} + \vec{F}_g$$

$$\vec{F}_{net} = m\vec{a}$$

$$\vec{a} = \frac{\vec{F}_{net}}{m}$$

$$\vec{a} = \frac{\vec{F}_{thrust} + \vec{F}_g}{m}$$

$$x = x_0 + V_0 t + \frac{1}{2}\vec{a}t^2$$

$$x = 0 + 0 + \frac{1}{2}\frac{\vec{F}_{thrust} + \vec{F}_g}{m}t^2$$

$$x = \frac{1}{2}\frac{\vec{F}_{thrust} + \vec{F}_g}{m}t^2$$

$$= \frac{1}{2} \cdot \frac{2.00 \times 10^5 N + 1.40 \times 10^4 kg \cdot 9.81 m/s^2}{1.40 \times 10^4 kg} \cdot 60 s^2$$

$$= \frac{1}{2} \cdot \frac{2.00 \times 10^5 N - 1.37 \times 10^5 N}{1.40 \times 10^4 kg} \cdot 60 s^2$$

$$= \frac{1}{2} \cdot \frac{6.26 \times 10^4 N}{1.40 \times 10^4 kg} \cdot 3600 s^2$$

$$= 4.47 \times 10^4 m/s^2 \cdot 1.800 \times 10^3 s^2$$

$$= 8.046 \times 10^7 m$$

#8.re.

$\vec{F_{g_1}} = (m_A + m_B)g$
$= .5kg \cdot -9.81 \, m/s^2$
$= -4.905N$

$\vec{F_{g_2}} = m_B g$
$= .3kg \cdot -9.81 \, m/s^2$
$= -2.943N$

$\vec{F_{net_1}} = m_1 \vec{a}$
$\vec{F_{net_2}} = m_2 \vec{a}$

$m_1 = m_A + m_B = .5kg$ $\quad m_2 = m_B$

a) $\vec{F_{T1}} = \vec{F_{net_1}} - \vec{F_{g_1}}$
$= m_1 \vec{a} - m_1 g$
$= 0 - -4.905N$
$= 4.905N$

$\vec{F_{T2}} = \vec{F_{net_2}} - \vec{F_{g_2}}$
$= m_2 \vec{a} - m_2 g$
$= 0 - -2.943N$
$= 2.943N$

b) $\because V_{avg} = +5 m/s$
$\therefore a = 0$
$\vec{F_{T1}} = 4.905N$ $\quad \vec{F_{T2}} = 2.943N$

c) $\vec{F_{T1}} = \vec{F_{net_1}} - \vec{F_{g_1}}$
$= m_1 \vec{a} - -4.905N$
$= .5kg \cdot 2 \, m/s^2 + 4.905N$
$= 1N + 4.905N$
$= 5.905N$

$\vec{F_{T2}} = \vec{F_{net_2}} - \vec{F_{g_2}}$
$= m_2 \vec{a} - -2.943N$
$= .3kg \cdot 2 \, m/s^2 + 2.943N$
$= .6N + 2.943N$
$= 3.543N$

d) $\vec{F_{T1}} = \vec{F_{net1}} - \vec{F_{g1}}$

$= m_1\vec{a} - -4.905N$

$= .5kg \cdot -2 m/s^2 + 4.905N$

$= -1N + 4.905N$

$= 3.905N$

$\vec{F_{T2}} = \vec{F_{net2}} - \vec{F_{g2}}$

$= m_2\vec{a} - -2.943N$

$= .3kg \cdot -2 m/s^2 + 2.943N$

$= -.6N + 2.943N$

$= 2.343N$

e) $\vec{F_{T1}} = \vec{F_{net1}} - \vec{F_{g1}}$

$\vec{F_{T1}} = m_1\vec{a} - \vec{F_{g1}}$

$m_1\vec{a} = \vec{F_{T1}} + \vec{F_{g1}}$

$\vec{a} = \frac{\vec{F_{T1}} + \vec{F_{g1}}}{m_1}$

$= \frac{10N + -4.905N}{.5kg}$

$= 5.095N / .5kg$

$= 10.19 m/s^2$

#1 re

$V = \frac{2\pi r}{T}$ $\quad a = \frac{v^2}{r}$

a) $100 km/h = 27.78 m/s$

$a = \frac{27.78 m/s^2}{50.0 m}$

$= 15.43 m/s^2$

$= 1.572g$

b) $1.50 \times 10^3 km/h = 416.67 m/s$

$5.00 km = 5000 m$

$a = \frac{416.67 m/s^2}{5000 m}$

$= .083 m/s^2$

$= .00849 g$

c) $V = \frac{2\pi \cdot 1.5m}{.5s}$

$= 18.85 m/s$

$a = \frac{18.85 m/s^2}{1.5m}$

$= 236.87 m/s^2$

$= 24.15g$

d) 33.1 rpm = 1.813 spr (T)

6.0 in = .15 m

$V = \frac{2\pi r}{T}$

$= \frac{2\pi \cdot .15\,m}{1.813\,s}$

$= .520\ m/s$

$a = \frac{V^2}{r}$

$= 1.802\ m/s^2$

$= .183g$

e) 30,000 rpm = .002 spr (T)

15.0 cm = .15 m

$V = \frac{2\pi r}{T}$

$= \frac{2\pi \cdot .15\,m}{.002\,s}$

$= 471.24\ m/s$

$a = \frac{V^2}{r}$

$= \frac{471.24\ m/s^2}{.15\,m}$

$= 1{,}480{,}440.66\ m/s^2$

$= 150{,}911.38g$